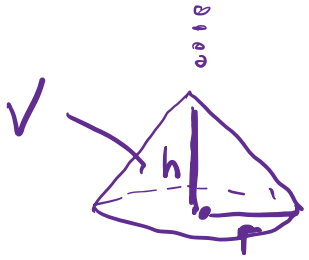
Problem 9. TIER 2 — RECALL A GEOMETRIC FACT

Gravel falls at 30 ft^3/min onto a conical pile whose **height always equals its base radius**. How fast is the height rising when the pile is 10 ft high?

(1) 

$r = h$

(2) $\frac{dV}{dt} = 30 \frac{\text{ft}^3}{\text{min}}$

$\frac{dh}{dt} = ?$, $h = 10$

1. Draw + Label
2. What & when
3. Equation
4. Differentiate w.r.t. t
5. Sub + Solve

(3) 

$V = \frac{1}{3} B h$

$V = \frac{1}{3} \pi r^2 h$

h

$V = \frac{1}{3} \pi h^2 h \rightarrow V = \frac{1}{3} \pi h^3$

(4) $\frac{d}{dt} V = \frac{d}{dt} \frac{1}{3} \pi h^3$

$\frac{dV}{dt} = \frac{1}{3} \pi 3 h^2 \frac{dh}{dt}$

(5) $30 = \pi 10^2 \frac{dh}{dt}$

$\frac{dh}{dt} = \frac{30}{100\pi} = \frac{3}{10\pi} \frac{\text{ft}}{\text{min}}$

$\approx .0955 \frac{\text{ft}}{\text{min}}$